CIEM4000-26
Fluid-Structure-Interaction Group Assignment 2026-2027

General introduction

This group assignment is compulsory and determines the final grade for the FSI unit of CIEM4000-26. It consists of several questions, which are all introduced in one of the seven workshops. It consists of three parts, on: I) Fluid dynamics (waves and flow), II) Loads induced by waves and currents, and III) Fluid-structure interaction. The group assignment can be made in groups of minimum 3 and maximum 4 students. You can register your group via Brightspace. Important dates and deadlines are given in Table 1.

Table 1 - Dates and deadlines 2026-2027.

No.	Date	Part	Topic to work on (advised)	Recommended activity
1.	12-11-2026	I	Part I, questions 1,2	-Assignment Part I is introduced. -First opportunity to ask your questions in person. -Form groups. Get to know the assignment.
2.	19-11-2026	I	Part I, questions 3,4	**Deadline** for registration of your group.
3.	26-11-2026	I	Part I, questions 5-8	
4.	03-12-2026	II	Tour Deltares facilities Guest lecture on offshore bed protections	Assignment Part II is introduced in lectures this week
5.	10-12-2026	II	Entire Part II	
6.	17-12-2026	III	Part III, questions 1-3	Assignment Part III is introduced
7.	06-01-2027	III	Part III, question 4	
	15-01-2027		**Deadline (18.00h)** of final assignment report	

It is recommended to start working on the assignment in week 2.1 (first workshop) and make use of the in-person workshops to ask all your questions. You need time to finish the assignment. The time in the workshops is mainly intended for understanding the assignment and asking questions. You need to do the work (self-study) in the time normally allocated to studying for the exam. Note that the workshops differ to those of the CIEM4000-26 sub-unit Mechanics of Slender Structures, where the work is done in the class, and self-study time is allocated to study for the exam (the FSI sub-unit does not have an exam).

The schedule in Table 1 states the topics that will be dealt with in each workshop. At the indicated times, it is certain that an expert on that part of the sub-unit will be present. The grading scheme is included in the table in Appendix A.

Assignment description

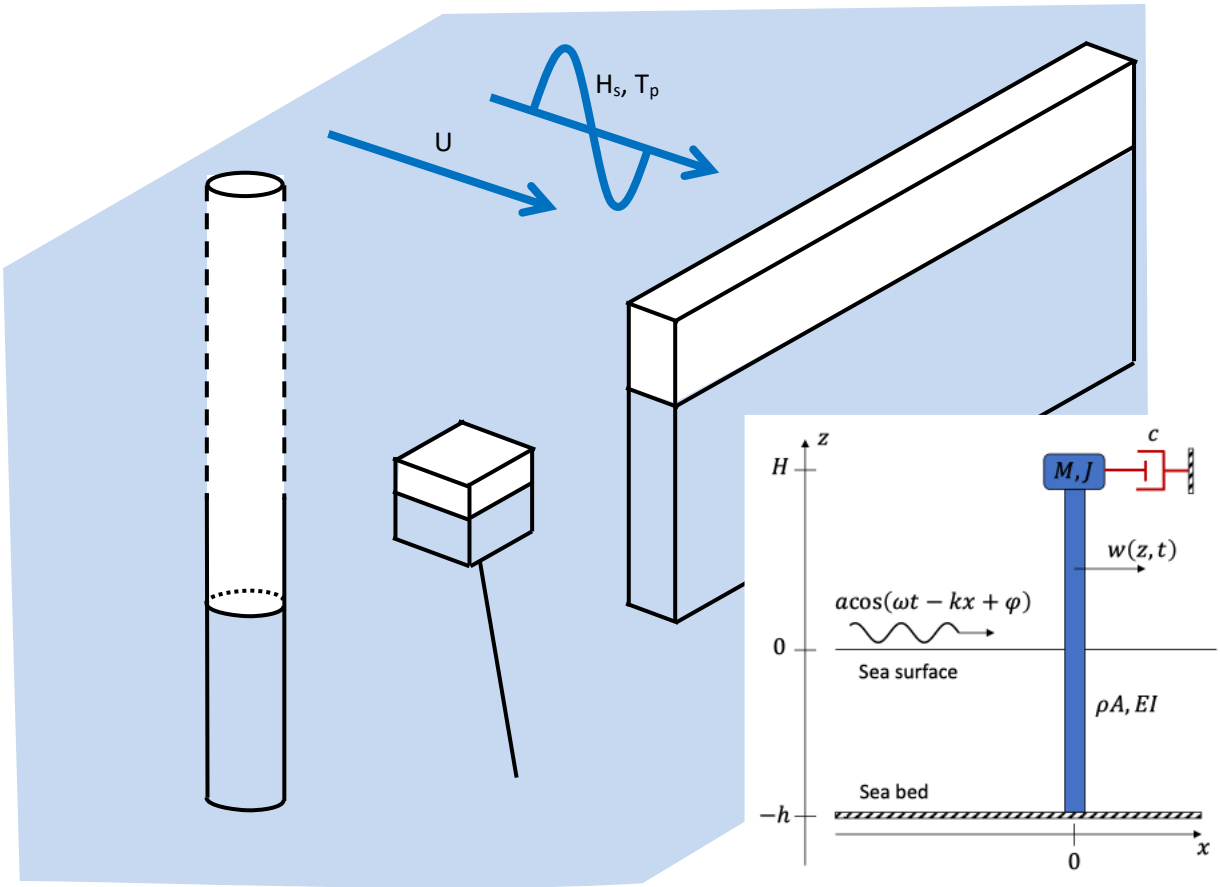


Figure 1 – The highly schematized offshore and hydraulic structure you will be working on.

You will work on a schematized offshore structure (a monopile) a hydraulic structure (a vertical wall, e.g. as in flood gates or caisson breakwaters) and on the dynamics of a mooring line in sheared current, as depicted in Figure 1. The questions follow the structure of the lectures, with the same parts on I) waves and flow, II) loads, and III) response (fluid-structure interaction).

Some input parameters are dependent on (the sum of) your student numbers and given in Table 2 and Table 3.

Table 2 - Student number-dependent information.

Last digit of the sum of student IDs	**= 1,2,3**	**= 4,5,6**	**= 7,8,9,0**
Water depth (m)	20	25	30
Wave height, H_s (m)	6	7	8
Current velocity, *U* (m/s)	2	1.5	1
Pile diameter (m)	4.5	5.5	6
Wall width, *W* (m)	60	70	80

Table 3 – Additional information for Part III.

Last digit sum of student IDs	**= 1,2,3**	**= 4,5,6**	**= 7,8,9,0**
Tower height above water (H)	85 m	105	120 m
Monopile Wall thickness (t)	100 mm	100 mm	100 mm
Rotor-Nacelle-Assembly mass (M)	350 tons	450 tons	675 tons
Aerodynamic damping (c)	30 kN s m-1	35 kN s m-1	40 kN s m-1

Reporting

Each group must hand in one clear and concise report (max. 25 pages[1]) with their results. Additional code and figures which support your work can be included in an appendix, though the main results and answers to the questions in the present document need to be provided in the 25-page report. The document must also be understandable and complete without reading the appendices. In case of an incomplete or incorrect answer, we will consider the appendix in determining the grade for a specific question. The report must be digitally submitted via Brightspace. Submissions up to 3 days after the deadline will be taken into consideration, but one point will be subtracted from the final grade.

Please provide clear answers to the questions including the method you used, derivations or sources of the equations you used, your assumptions, values you implemented, final plots, and a concise discussion of those. (Note that copying in a Python script does not count as a sufficient explanation of your method). You need to explicitly assume characteristic values for the unknown parameters. Discuss which uncertainties and assumptions are largest in your approach, and potential mistakes you made in case you have doubts about the results obtained. It is not strictly necessary to share your developed computer code(s)/models, but you may be asked to in case of suspicions of plagiarism.

AI use

You may use AI to assist with specific tasks such as drafting (limited length) texts or code, refining and evaluating your work. You need to report in an appendix how AI was used in the assignment, both for text and coding separately.

You must critically evaluate and modify any AI-generated content you use. You may think of AI as a colleague you cannot quite trust and who must be checked all the time:

1. Before: In your report, you will critically examine and justify any assumptions you make before starting a calculation.
2. During: You must be able to explain all parts of your calculation or code when queried, including those parts where AI helped you.
3. After: Once you have obtained your result (e.g., the numerical answer to a question), you must examine its validity and demonstrate this in your report.

You are finally responsible for the content in the report, explaining the methods and input used, schematizations applied, and verification of the results.

Word of advice: you will be a more proficient AI user when you have done the work (coding)

yourself at least once! Write your own code now and you will reap the benefits throughout your career!

[1] Based on normal margins and font (12-point Calibri, regular line spacing).

Description

Part I – Fluid dynamics

Flow

1. Unobstructed flow: Using a set of measurements that can be found here: https://surfdrive.surf.nl/s/9G3XdNay8GQRyL9,
    a. fit the log-profile through the measurements
    b. determine the shear stress in two ways: from fitting the log-profile and from extrapolation of the Reynolds stress to the bed (from the log-fit you can also obtain the roughness height of the bed).
    c. discuss the difference.
2. Obstructed flow: Sketch the streamlines of a spatially uniform current around a wall (2-dimensional top view, the wall has a width *W*). The current approaches the wall in the direction normal to the width of the wall. Indicate the following items:
    a. Separation points, stagnation point, mixing layer, wake, region of highest velocity, region of highest turbulence intensity.
    b. Indicate the region where you expect the bed protection to be subject to the greatest stresses.

Waves

3. Plot a (variance density) spectrum that has the value of H_s given in the table. Choose a suitable spectral shape and peak period for locally generated (and thus steep) wind waves. You can assume long-crested waves.
    a. Determine T_m, T_{m-10}, and T_{m01}.
    b. Calculate the wave steepness.
4. Create a random-phase time series of the wave field ($\eta(t)$ and $u(z,t)$) at a spatial location (x, y), for a given *z*, which can you can set to zero here (*z=0*), and in the absence of a structure.
    a. First, assume linear wave theory applies.
    b. Second, add second-order super-harmonic bound waves to your time series (model).
    c. Third, explain the role of sub-harmonic bound waves and how you might model them.

    - Explain the chosen frequency discretization and other assumptions.
    - Describe how the model was verified and validated.

    This model will be used in Parts II and III.
5. Compare the crest elevations (and wave heights) of the individual waves in your time series and compare to the Rayleigh distribution.
6. Determine the statistics of (you may use linear wave theory):
    a. velocity near the bed, and maximum pressure gradient near the bed from the time series.
    b. velocity near the surface
7. Discuss both a numerical model you could use to calculate the forces on the wall and a model you would not use for this purpose.

Part II – Loads

Unless stated otherwise, the Loads questions deal with combined wave-current load.

1. Determine the time-varying force on the pile using the Morison approach, and make a plot for the time around the largest force. Plus give (exceedance) distribution of peak forces (bonus: compare to Rayleigh).
2. Determine the standing wave load on the wall (using linear wave theory). Neglect the steady current.
3. In the middle of the wall, determine the wave overtopping discharge.
4. If a wide deck would be added to the wall, protruding 1 m into the sea at an elevation of 10 m above the water line, calculate the force impulse of the impulsive confined wave impact on this deck.
5. Determine either:
   - the minimum stable armour rock size needed to protect the bed around the monopile, or
   - the minimum stable armour rock size needed to protect the bed around the wall.
6. Determine the rock gradings for a (multi-layer) closed granular filter layer underneath the armour. Discuss whether the armour rock calculated under the previous question is still stable?
7. For a single layer open filter (only applying a double layer of the armour rock that you determined) determine the filter velocity near the sand-rock interface without the structure present. (you can assume Darcy flow in the rock material).
8. A test facility (wide flume) is available to test the bed protection design based on your calculated armour rock diameter (you can choose between wall or pile). It has a water depth up to 1 m deep with $H_{s,max}$ = 25 cm. Describe the required test setup (dimensions, setup, and what to measure) to validate the stability of (the top layer of) the bed protection.

Part III – Fluid-structure interaction

1. Determine the **steady-state response** of the pile to the wave field determined in Question 4 or Part I disregarding the aerodynamic damping. Provide graphs of:
   a. the steady-state displacement of the wind turbine at the top of the turbine (z = H) and
   b. of the bending moment at the cross section of the monopile that is most prone to fatigue.

   It is advisable to linearize the problem and use the superposition principle, i.e. derive the turbine response to the irregular wave train as the superposition of the responses to the regular waves that compose the wave train.

2. Account for the **aerodynamic damping** due to turbine operation which is accounted for by the linear dashpot with coefficient c as shown in Figure 1. Provide graphs of the same quantities as in question 1 and comment on the differences.

3. Determine the frequency- and depth- dependent added mass and added damping coefficients for the fundamental mode of the cylinder, with the wall thickness and diameter defined in Table 2 and its vibrational shape given as:
   $W(z) = W_0 \sin\left(\frac{\pi(z+h)}{2(h+H)}\right)$.
   You are asked to:
   a) Derive the z and ω (frequency)-dependent added mass and added damping experienced by the submerged part of the monopile.
   b) Plot the contour plots (in the $\omega - z\ plane$) of the added mass and added damping for 2 values of the water depth, namely the value from Table 1, and $h = 70\ m$.

4. Investigate the dynamic response of a vertical tensioned ($T = 30$ kN) mooring line placed in a uniformly distributed along the water depth current whose velocity is horizontal and constant in time (Figure 1). Specifically, you are asked to determine the response amplitude in a condition where vortex-induced vibrations develop.
   The modal shape of the mooring line is given as $W(z) = W_0 \sin\left(\frac{\pi z}{L}\right)$.
   For this purpose, you are to solve the following coupled differential equation:

$$\ddot{q} + \varepsilon(q^2 - 1)\dot{q} + q = A\ddot{y}$$

$$\ddot{y} + 2\zeta\Omega_n\dot{y} + \Omega_n^2 y = \frac{\rho D^2 L}{(m + m_a)8\pi^2 \mathrm{St}^2}\left(-2\pi \mathrm{St}\dot{y}\hat{C}_{x0}^0 + \frac{\hat{C}_{y1}^0}{\hat{q}_1}q\right)\sqrt{1 + 4\pi^2 \mathrm{St}^2\dot{y}^2}$$

   You can find more details about the equation and tuning parameters in the paper by Ogink and Metrikine (2010) available on Brightspace and online.

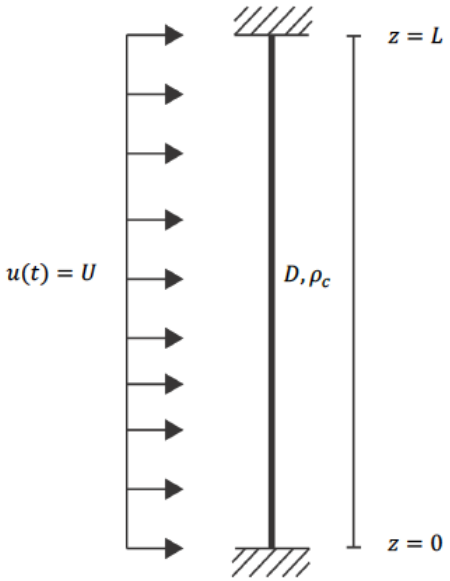


Figure 1 - a vertical tensioned string placed in a uniformly distributed current.

You are asked to:

a) determine the relevant parameters m, m_a and Ω_n for representing the mooring line as an equivalent SDOF system required to solve the coupled differential equation considering the shape of oscillation of the mooring line as provided.

b) Define the governing system of equations.
c) Determine (based on VIV theory) a constant current speed U which you expect VIV to develop for this system.
d) Confirm that VIV develops by solving the system of equations and plotting the amplitude of oscillation of the mooring line.

Table 4 - Parameters and their values to be used in this assignment.

Parameter	Value
Cable length (L)	50 m
Cable diameter (D)	0.5 m
Water density (ρ)	1025 kg/m3
Damping ratio (ζ)	2% of critical
Cable density (ρ_c)	7880 kg/m3

References

[1] Schiereck and Verhagen (2019) Bed, bank and shore protection. 2nd edition, revised 2019. Delft Academic Press. https://resolver.tudelft.nl/uuid:df203dde-6cca-4375-85d2-8b2dbce613cb

[2] Ogink, R. H. M., and Metrikine, A. V. (2010). A wake oscillator with frequency dependent tuning coefficients for the modeling of vortex-induced vibration. J. Sound. Vib. 329 (26), 5452–5473. https://www.doi.org/10.1016/j.jsv.2010.07.008

Appendix A – Grading scheme

Below the expected content of the report is given, including the maximum score per item.

Table 3 - Grading scheme.

Knock out criteria	
Report pages < 25 (excluding cover)	If limit is exceeded the report will not be checked.
Part 1 (40%)	**Flow and waves**
1 (5%)	•
2 (5%)	•
3 (5%)	•
4 (10%)	•
5 (5%)	•
6 (5%)	•
7 (5%)	•
	•
Part 2 (30%)	**Loads**
1 (5%)	•
2 (2.5%)	•
3 (2.5%)	•
4 (5%)	•
5 (5%)	•
6 (2.5%)	•
7 (2.5%)	•
8 (5%)	•

Part 2 (30%)	**FSI**
1 (5%)	•
2 (5%)	•
3 (5%)	•
4a (2.5%)	•
4b (2.5%)	•
4c (5%)	•
4d (5%)	•